Intersections of Hyperplanes

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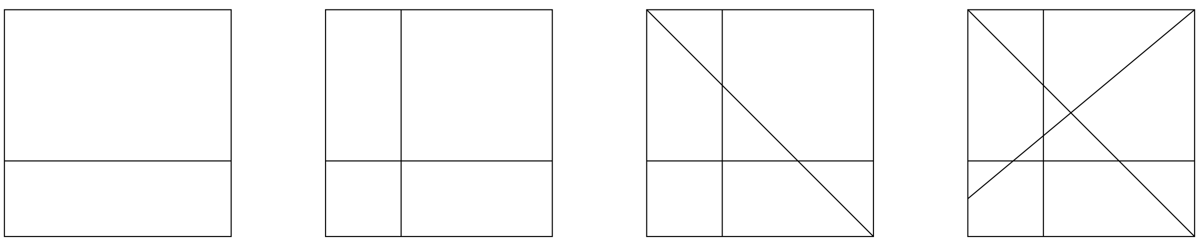
2 Dimensions

Question 1. *Given b lines in the plane, what is the maximum number of regions that they can divide the plane into?*

We can figure this out for some small values of b:



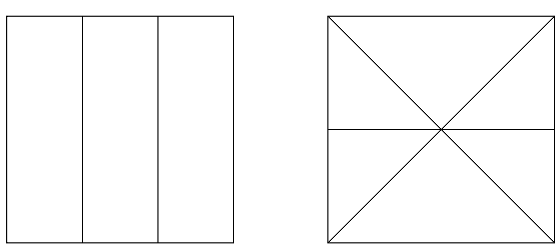
$b = 1 : 2$ $\quad$ $b = 2 : 4$ $\quad$ $b = 3 : 7$ $\quad$ $b = 4 : 11$

If we look at the difference between these values, we see that they are 2, 3, and 4 respectively. We thus make the following conjecture.

Conjecture 1. *The answer to Question 1 is* $1 + \big(1 + 2 + \cdots + b\big) = 1 + \frac{b(b+1)}{2}$

We will prove that this is an upper bound for the number of regions first. Let us try to prove this by induction, with the base case of $b = 1$ being clear. Suppose that this is an upper bound for b lines, and that we are given $b+1$ lines. Fixing a line and removing it, the number of regions created by the remaining b lines is at most $1 + \frac{b(b+1)}{2}$. Consider the number k of distinct points that our removed line intersects the other lines. The line passes through $k + 1$ regions, and so creates $k + 1$ new regions. Since $k \leq b$, we get that the line creates at most $b+1$ new regions, and so we have an upper bound of $1 + \frac{b(b+1)}{2} + (b+1) = 1 + \frac{(b+1)(b+2)}{2}$.

Now, it remains to show that this value is actually achievable. However, notice that do not *always* get the value described above. For example, consider the following examples:



What are the problems here? In the first, the two lines are parallel, and their lack of intersection creates less regions. In the second, the 3 lines are concurrent, and this once again creates less regions. However, if we do not have these problems, it seems that the maximum is achieved.

Conjecture 2. *Conjecture 1 is true, and occurs when no* 2 *distinct lines are parallel and no* 3 *distinct lines are concurrent.*

How would we go about proving this? One method is to use induction. In the base case of $b = 1$, these conditions vacuously hold, and we can achieve the value of Conjecture 1. Suppose that Conjecture 2 holds for b lines, and that we have $b+1$ lines satisfying the conditions of Conjecture 2. Fixing a line and removing it, we get b lines satisfying the conditions, and so there are $1 + \frac{b(b+1)}{2}$ regions. Now, suppose that the line we removed intersects the other lines at k distinct points. Then, it passes through $k+1$ of the regions that they create, and so adds $k+1$ new regions when added back in. Since we have the conditions of Conjecture 2, we get that $k = b$, and so the number of regions we have is $1 + \frac{b(b+1)}{2} + (b+1) = 1 + \frac{(b+1)(b+2)}{2}$. Hence, we have shown that the conditions of Conjecture 2 imply the value of Conjecture 1.

Notice that the method we used shows why we must require that no 2 distinct lines are parallel and no 3 distinct lines are concurrent. If the line we had removed were parallel to another line, they would not intersect, and so k would be at most $b-1$. Similarly, if the line were concurrent with 2 other lines, there would only be 1 distinct intersection point, meaning that k would be at most $b-1$.

Now, we must show that it is always possible to draw lines satisfying these conditions. Once again, we will use induction on b, with $b = 1$ being trivial. If we can draw b lines satisfying the conditions, create a $b+1$st line not parallel to any of these b lines, and shift it so that it does not go through any of the finitely many intersection points of these b lines.

Hence, we have shown both that our value is an upper bound and that it is achievable, so it is the maximum number of regions that can be created.

Solution. The maximum number of regions that can be created by b lines in the plane is $1 + \frac{b(b+1)}{2}$, and this happens with no 2 distinct lines are parallel and no 3 distinct lines are concurrent.

3 Dimensions

Question 2. *Given b planes in space, what is the maximum number of regions that they divide the space into?*

Notation. We will write $R_2(b)$ for the maximum number of regions that b lines divide the plane into, and $R_3(b)$ for the maximum number of regions that b planes divide space into.

Once again, we can compute some small values of b:

b	$R_3(b)$
1	2
2	4
3	8
4	15
5	26

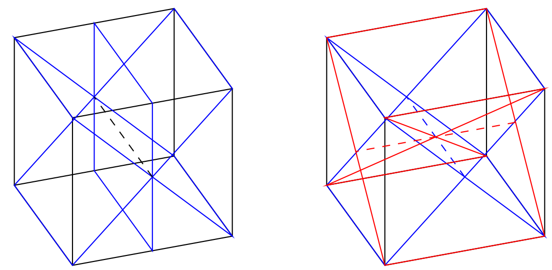
It seems that $R_3(b+1) - R_3(b) = R_2(b)$. With this, we make the following conjecture:

Conjecture 3. *The solution to Question 2 is*

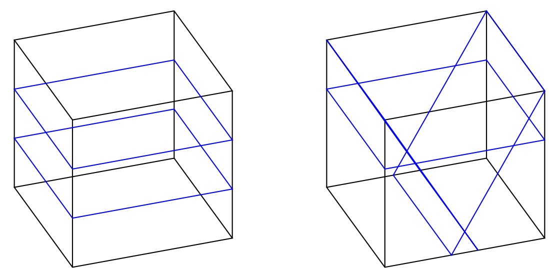
$$2 + \sum_{i=1}^{b-1} \left(1 + \frac{i(i+1)}{2}\right) = \frac{b^3 + 5b + 6}{6}.$$

We can prove that this is an upper bound by induction on b, with $b = 1$ being trivial. If this value is an upper bound for b planes, consider $b+1$ planes. Fix a plane and remove it, to get that the remaining b planes divide space into at most $\frac{b^3+5b+6}{6}$ regions. Consider the regions that this fixed plane is divided into by its intersections with the other b planes. These intersections create b lines, and so this plane is divided into at most $1 + \frac{b(b+1)}{2}$ regions. Thus, it intersects at most $1 + \frac{b(b+1)}{2}$ regions in space, and creates at most those many new ones. Hence, the number of regions created by the $b+1$ planes is at most $\frac{b^3+5b+6}{6} + 1 + \frac{b(b+1)}{2} = \frac{(b+1)^3+5(b+1)+6}{6}$, and we have an upper bound.

Once again, we must show that the value is achievable. Inspired by the 2-dimensional case, we require that no 3 planes intersect in the same line, and that no 4 planes intersect in the same point, to avoid cases like below.



We also require that no 2 planes are parallel, and that any 3 planes intersect in a point, to avoid cases like below.



Conjecture 4. *Conjecture 3 is true, and occurs when no* 2 *planes are parallel, any* 3 *planes intersect at a point, no* 3 *planes intersect at in a line, and no* 4 *planes intersect in a point.*

Let us verify this. By the line of reasoning above, it comes down to verifying that the intersections of a fixed plane with each other plane satisfies the conditions we established for 2 dimensions. First, no 2 planes are parallel and no 3 planes intersect in a line, meaning our fixed plane intersects each of the other b planes in a unique line. Since any 3 planes intersect at a point, no 2 of these lines are parallel. Since no 4 planes intersect in a point, no 3 of these lines are concurrent. Hence, the conditions are satisfied, and so we can achieve the upper bound.

Solution. The maximum number of regions that can be created by b planes in space is $\frac{b^3+5b+6}{6}$, and this happens when: 2 planes always intersect, 3 planes always intersect in exactly a point (not a line), and 4 planes never intersect in a point.

d Dimensions

We'll now turn to the following, more general question:

Question 3. *Given b ($d-1$-dimensional) hyperplanes in $\mathbb{R}^d$, what is the maximum number of regions that they divide space into?*

Notation. We'll write $R_d(b)$ for the maximum number of regions that b hyperplanes can divide $\mathbb{R}^d$ into.

We would expect, from our low-dimensional examples, that $R_{d+1}(b+1) - R_{d+1}(b) = R_d(b)$. Let's figure out some base cases, so that we can use this to give a proper answer. We know that $R_d(0) = 1$, since 0 hyperplanes leaves 1 region in $\mathbb{R}^d$ (namely, the whole space). We can also consider the "degenerate" case of $R_0(b)$, which must be 1, since it is impossible to draw a -1-dimensional hyperplane in $\mathbb{R}^0$. With this, we get the following:

Conjecture 5. *The solution to Question 3 is $R_d(b)$, where $R_0(b) = R_d(0) = 1$ and $R_{d+1}(b+1) - R_{d+1}(b) = R_d(b)$.*

We can see that $R_1(b) = b + 1$, which makes sense, since b points divides a line into $b + 1$ segments. We can also compute that this agrees with our previously computed values of $R_2(b)$ and $R_3(b)$. We'll prove by induction on d, having just verified the base case, that this recurrence provides an upper bound.

Suppose that the recurrence-defined $R_d(b)$ is an upper bound of the solution to Question 3 for all b. We'll show that $R_{d+1}(b)$ is an upper bound by induction on b. The base case, $b = 0$, is true by definition. Suppose that $R_{d+1}(b)$ is an upper bound on the number of regions formed, and consider $b+1$ hyperplanes. Removing one of the hyperplanes, we get at most $R_{d+1}(b)$ regions. Consider the intersections of these b hyperplanes with the hyperplane we removed. These divide the hyperplane into $k \leq R_d(b)$ regions. Each of these corresponds to adding a new region in our original space, so we see that there are at most $R_{d+1}(b) + R_d(b) = R_{d+1}(b+1)$ regions in our original space, as desired.

How would we achieve equality? We basically need to avoid "redundant" intersections (e.g. 3 concurrent lines), while still enforcing intersections (e.g. no parallel lines). We can encapsulate this by requiring that any n hyperplanes intersect in exactly $d - n$ dimensions for $2 \leq n \leq d+1$ (where intersecting in -1 dimensions means not intersecting).

Conjecture 6. *Conjecture 5 is true, and occurs when all collections of n hyperplanes intersect in exactly $d - n$ dimensions for $2 \leq n \leq d+1$, where intersecting in -1 dimensions means not intersecting.*

Verifying this requires showing that the conditions in $d+1$ dimensions force the conditions in d dimensions when examining the intersection of a fixed hyperplanes with b other hyperplanes. If the conditions in $d+1$ dimensions are satisfied by $b+1$ hyperplanes, then in particular any 2 hyperplanes intersect in a $d-2$ dimensional hyperplanes, and so we find b $d-2$-dimensional hyperplanes cutting up our fixed hyperplane. Suppose we have a collection of n of these. We want to show that they intersect in $d-1-n$ dimensions. We know that the intersection of these n $d-2$-dimensional hyperplanes is the same as the intersection of n corresponding $d-1$-dimensional hyperplanes, along with our fixed hyperplane. Thus, the intersection is $d-(n+1) = d-1-n$ dimensional, as desired.

Let's try to find a closed form for this. We know that

$$R_0(b) = 1 \qquad R_1(b) = 1 + b \qquad R_2(b) = 1 + \frac{b(b+1)}{2} \qquad R_3(b) = \frac{b^3 + 5b + 6}{6}.$$

We can rearrange to find a nicer pattern:

$$R_0(b) = 1 \qquad R_1(b) = 1 + b \qquad R_2(b) = 1 + b + \frac{b(b-1)}{2} \qquad R_3(b) = 1 + b + \frac{b(b-1)}{2} + \frac{b(b-1)(b-2)}{6}.$$

It seems like

$$R_d(b) = \sum_{i=0}^{d} \binom{b}{i}.$$

If we verify that this satisfies the same recurrence and base cases, we are done. We can see that

$$\sum_{i=0}^{0} \binom{b}{i} = \binom{b}{0} = 1 = R_0(b)$$

and that

$$\sum_{i=0}^{d} \binom{0}{i} = \binom{0}{0} = 1 = R_d(0),$$

so that the base cases agree. Now,

$$\sum_{i=0}^{d+1} \binom{b+1}{i} - \sum_{i=0}^{d+1} \binom{b}{i} = \sum_{i=0}^{d+1} \binom{b+1}{i} - \binom{b}{i} = \sum_{i=1}^{d+1} \binom{b}{i+1} = \sum_{i=0}^{d} \binom{b}{i},$$

and so this is the same as $R_d(b)$. We therefore have the following.

Solution. The maximum number of regions that $\mathbb{R}^d$ can be divided into by b hyperplanes is $\sum_{i=0}^{d} \binom{b}{d}$, and this occurs when any n hyperplanes intersect in exactly $d - n$ dimensions ($2 \leq n \leq d+1$).